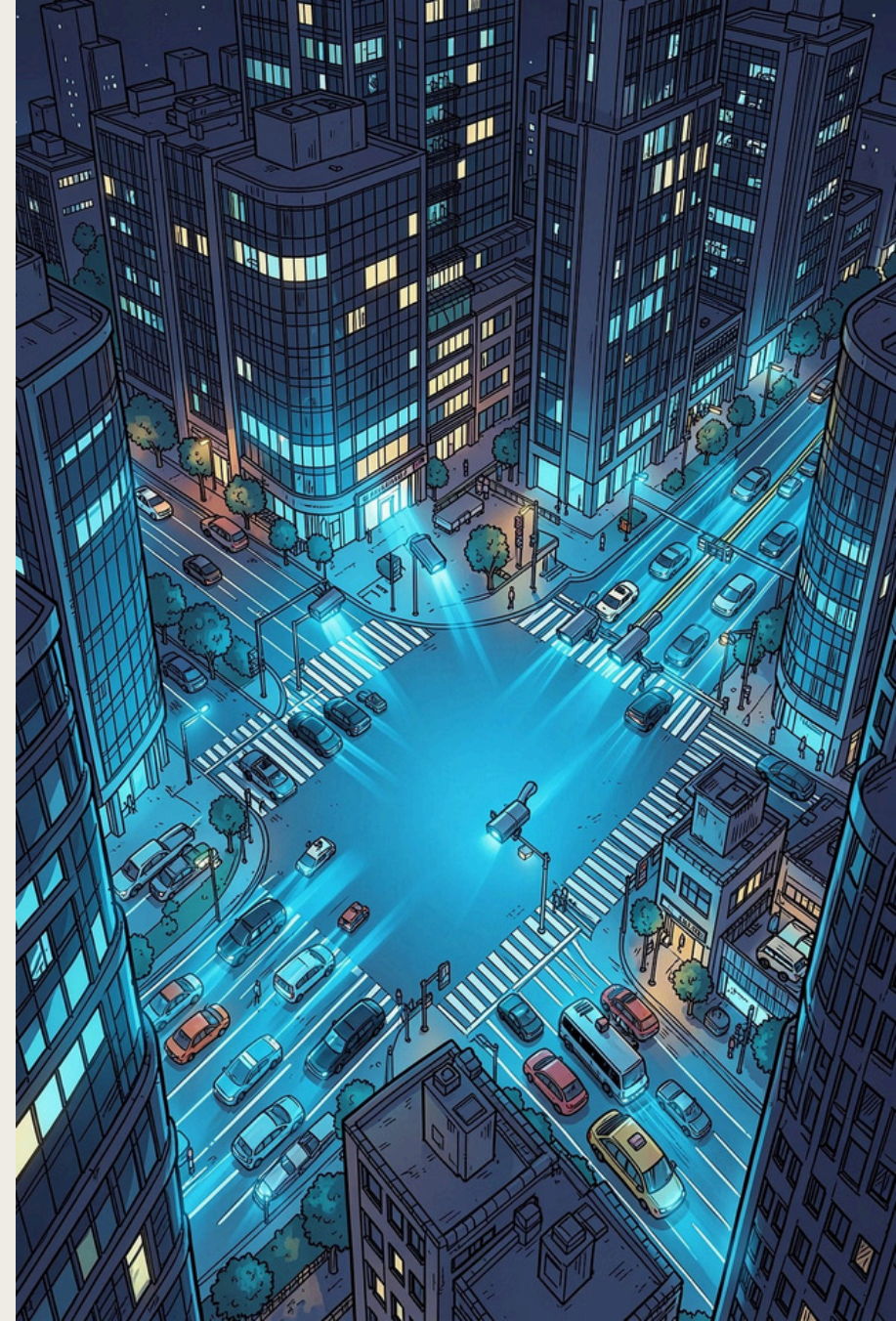


VisionGuard AI

Intelligent Traffic Violation Evidence & Threat
Intelligence System

Scalable, Explainable and Tamper-Resistant Traffic
Enforcement for Smart Cities

TEAM : Navora Flux



The Problem with Current Traffic Monitoring

Existing systems are reactive, opaque, and operationally fragile—unable to meet the demands of modern smart-city enforcement.

Manual Overload

Millions of images require human review — slow, costly, and error-prone at scale.

Environmental Blindness

Rain, night, motion blur, and shadows degrade detection accuracy with no adaptive compensation.

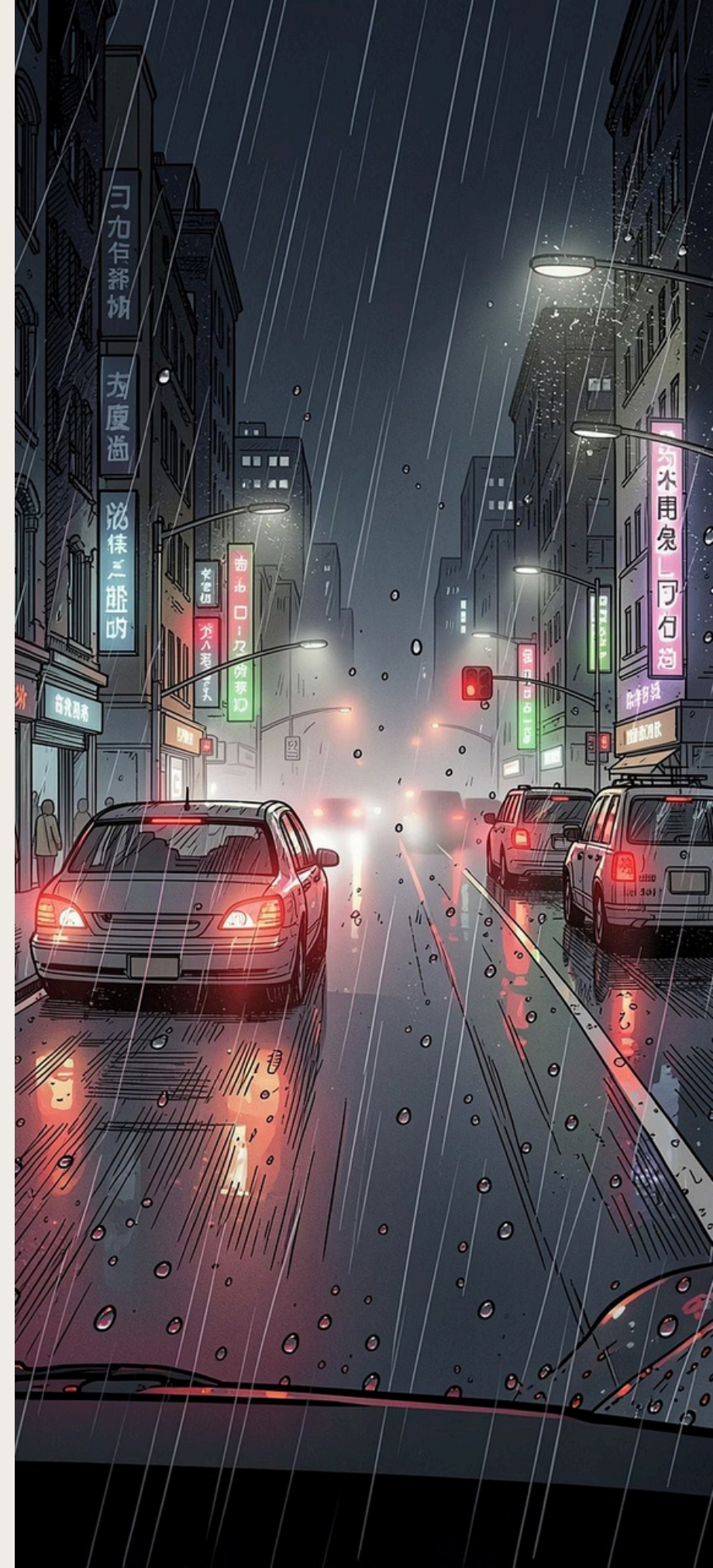
No Explainability

Black-box decisions cannot be legally defended or operationally audited.

No Threat Intelligence

Zero repeat-offender tracking, risk heatmaps, or officer deployment recommendations.

- ❑ **VisionGuard AI transforms passive camera networks into an intelligent, legally defensible evidence generation platform.**



Why VisionGuard is Different

Four pillars that set VisionGuard apart from basic YOLO detectors and single-class classifiers.



Environmental Awareness

Adaptive detection across rain, night, motion blur, and shadow conditions — no scenario left unhandled.



Explainable AI

Every decision includes a human-readable reason, confidence score, and severity rating — auditable by design.



Evidence Integrity

SHA256 hashing, GPS, timestamp, and camera ID create a tamper-proof chain of custody for every violation.



Threat Intelligence

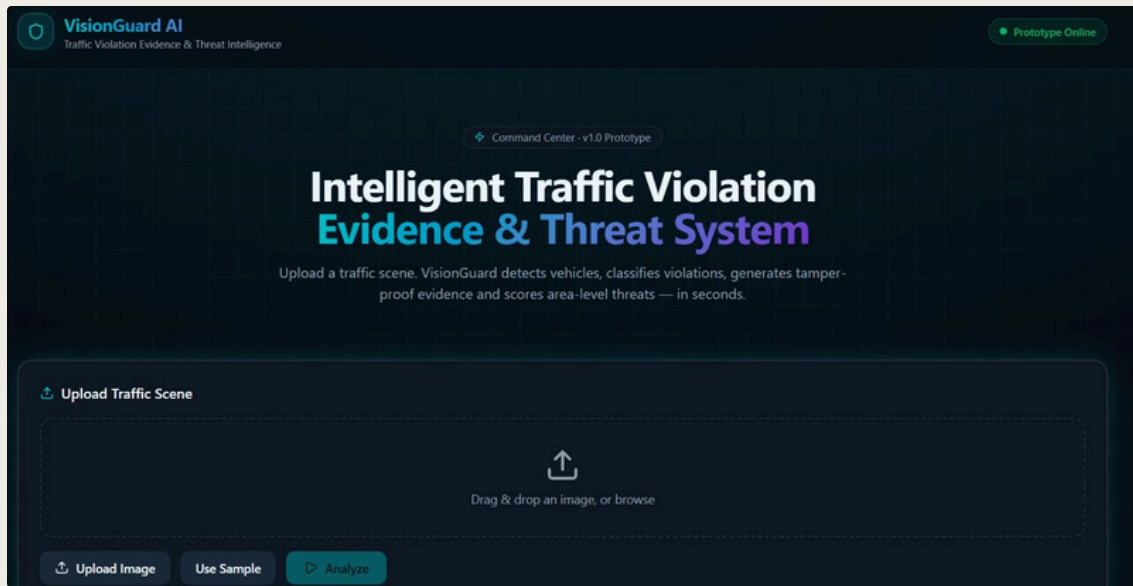
Repeat-offender tracking, risk heatmaps, and dynamic officer deployment recommendations.

VisionGuard AI Architecture

A nine-stage end-to-end pipeline — from raw camera feed to actionable intelligence — with built-in explainability and evidence integrity at every step.



Prototype — VisionGuard Command Center

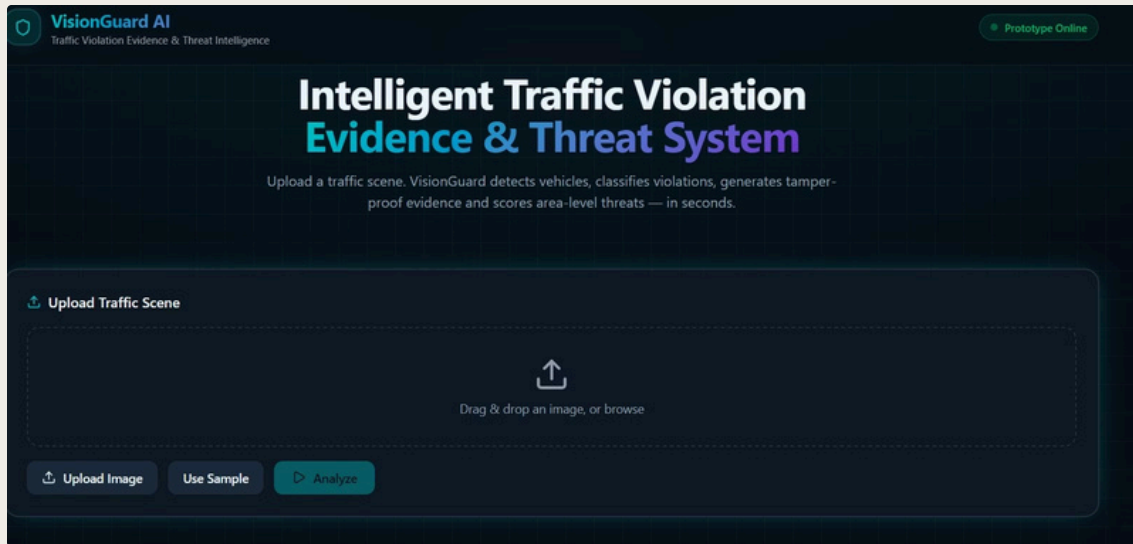


Hero Interface

The Command Center provides a unified upload interface with real-time analysis triggers — designed for both single-image review and batch processing workflows.

- Drag-and-drop image upload
- One-click Analyze trigger
- Live violation status panel
- Evidence preview with metadata overlay

Processing Pipeline — Intelligent Evidence Workflow



The pipeline orchestrates eight sequential stages — each producing structured, explainable output that feeds the next stage and ultimately the Command Center.

1. Environmental Awareness

2. Image Enhancement

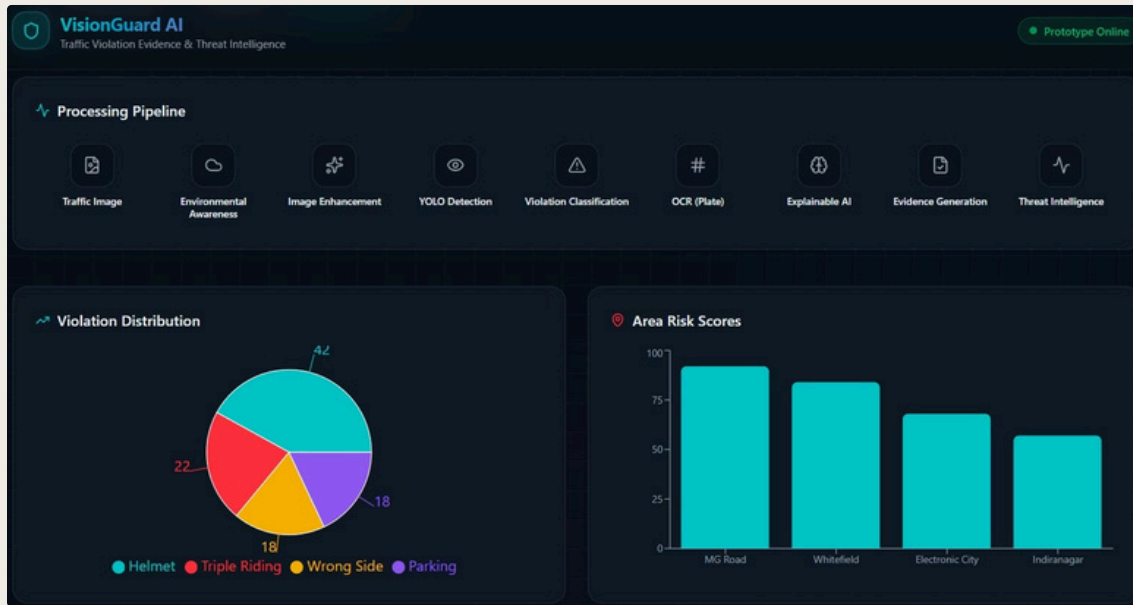
3. YOLO Detection

4. Violation Classification

5. OCR · Explainability · Evidence

6. Threat Intelligence

Analytics Dashboard



Violation Distribution & Area Risk Scores

The analytics module translates raw detections into operational intelligence — enabling data-driven officer deployment and policy decisions.

- **Violation Distribution** — categorical breakdown by type and severity
- **Area Risk Scores** — geospatial risk aggregation by zone
- **Threat Index** — composite score combining frequency and severity
- **Officer Assignment** — recommended patrol routing based on live data

Supported Violations & Datasets

Supported Violations

 **Helmet Violation**

 **Triple Riding**

 **Seatbelt Violation**

 **Wrong-Side Driving**

 **Illegal Parking**

 **Red Light Violation**

 **Stop Line Violation**

Training & Evaluation Datasets

- COCO — general object detection baseline
- IDD — Indian driving场景 diversity
- CCPD — Chinese license plate dataset
- Roboflow Helmet Dataset
- EasyOCR — plate recognition engine
- Synthetic Dataset — edge-case augmentation

Prototype Status & Roadmap

Current Prototype

- Concept Note & Architecture
- Command Center UI
- Explainable AI Design
- Evidence Integrity Framework
- Threat Intelligence Design
- Analytics Dashboard

Future Work

- Real-time CCTV integration
- Full YOLOv8 training pipeline
- Federated Learning for privacy-preserving edge training
- Edge AI deployment on traffic hardware
- E-Challan automation
- Smart City API integration

 Certain detections and analytics are simulated to demonstrate workflow orchestration and scalability under prototype-stage constraints.



VisionGuard AI

Transforming traffic cameras into explainable, intelligence-driven, legally defensible enforcement systems.

Environmental Awareness

Explainable AI

Evidence Integrity

Threat Intelligence

Prototype developed as a scalable solution framework for Automated Photo Identification and Classification for Traffic Violations Using Computer Vision.